National Institute of Technology Kurukshetra
Department of Computer Engineering

Problems Solving and Programming using C (CSIC101)

Mid-II Exam, Oct-2024

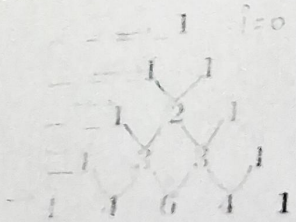
Timing: 50 Min. | Oct-2024, Semester-I | Max Marks: 20

1. Write a C program using a function named **leapyear(int X)** to display the leap years that contain the digit **'X'** given by the user, between two given year **n1** and **n2** both inclusive. For example, if **n1** and **n2** are 1990 and 2020 and if the digit **'X'** is 6, the outputs need to be: The number of leap years between 1990 and 2020 that contains the digit 6 are: 1996, 2016. **(5)**

2. Write a program that sums a sequence of integers. Assume that the first integer read with scanf() specifies the number of values remaining to be entered. Your program should read only one value each time scanf() is executed. A typical input sequence might be: **5** 100 200 300 400 500, where the **5** indicates that the subsequent five values are to be summed. **(5)**

3. Tower of Hanoi is a mathematical puzzle where we have three rods A, B, and C and n disks. The objective of the puzzle is to move the entire stack from rod A to another rod C where B can be used as an auxiliary rod, obeying the following simple rules:
   1) Only one disk can be moved at a time.
   2) Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack.
   3) No disk may be placed on top of a smaller disk.
   Your C program should print the precise instructions it will take to move the disks from the starting rods to the destination rod. For example, to move a stack of three disks from rod A to rod C, your program should print the following series of moves:
   Disk 1 from A to C
   Disk 2 from A to B
   Disk 1 from C to B
   Disk 3 from A to C
   Disk 1 from B to A
   Disk 2 from B to C
   Disk 1 from A to C **(5)**

4. A really interesting Number Patterns is Pascal's Triangle (named after Blaise Pascal, a famous French Mathematician and Philosopher).
   To build the triangle, start with "1" at the top, then continue placing numbers below it in a triangular pattern.Each number is the numbers directly above it added together. Write a C program and ask user to enter the number of rows: For Example :
   Enter the number of rows: 5



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